

SpectraTwin

Track IoT devices safely, even when their IPv6 address keeps changing

A3-PS007-TC170
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Problem

IoT devices keep changing address, so we lose track of them

- An IPv6 device changes its address often, about once a day.
- Our tools think it is a new device.
- Old logs and rules stop working.
- Hackers can hide in this gap.
- Tracking people to fix this is not okay. Privacy is lost.

~18.8B

IoT devices in the world (2024, estimate)

1 day

an IPv6 address often changes in about a day

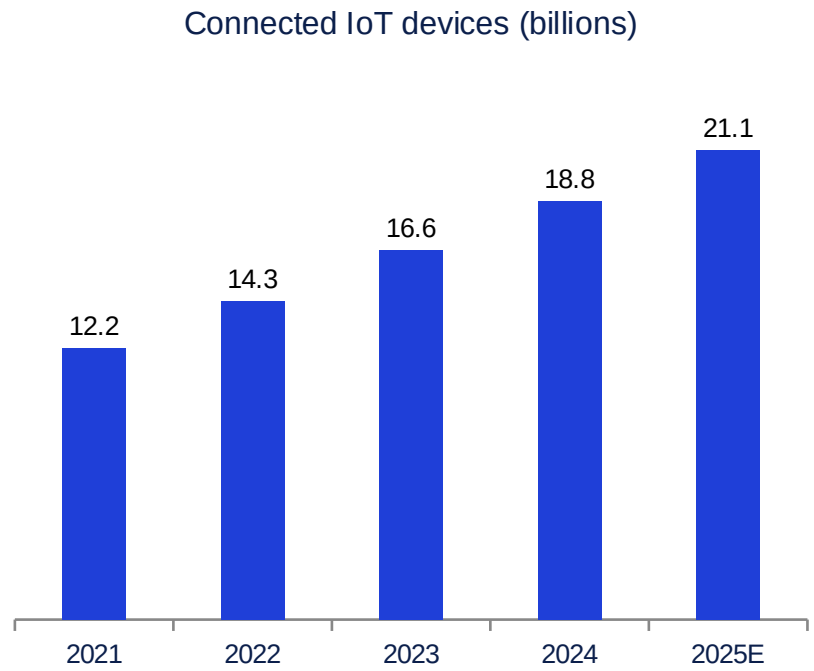
2 needs

keep the network safe and keep users private

Background

Why this problem is growing

- Most phones and laptops change their IPv6 address on their own [1].
- Every IoT device has habits: when it sends data, how big the packets are, which names it asks for [3].
- These habits stay the same when the address changes.
- India is using more IPv6, and privacy laws (DPDP Act 2023) are getting strict.



Source: IoT Analytics estimates. Please check before final submit.

Literature & RFC Review

What others did, and what is missing

RFC 8981 / 4941 [1]

Makes IPv6 addresses change, to stop tracking.

RFC 7721 [2]

Explains privacy risks in IPv6 addresses.

Sivanathan et al. [3]

Finds device type from its traffic habits.

SimCLR [4]

Learns patterns without labels.

What is missing: Old work either tracks devices (bad for privacy) or hides traffic (bad for monitoring). No one uses device habits, device place and privacy together.

Our Solution: SpectraTwin

A robot maps the place. We match devices by habit and place.



Why it is special (USP)

Place helps identity

Devices stay in one spot. So the place stays same when the address changes.

Fake device check

A "camera" that acts like a laptop gets a red flag.

Privacy we can measure

Hidden IDs, rough location, and a graph that shows the trade-off.

Improvement

Before and after

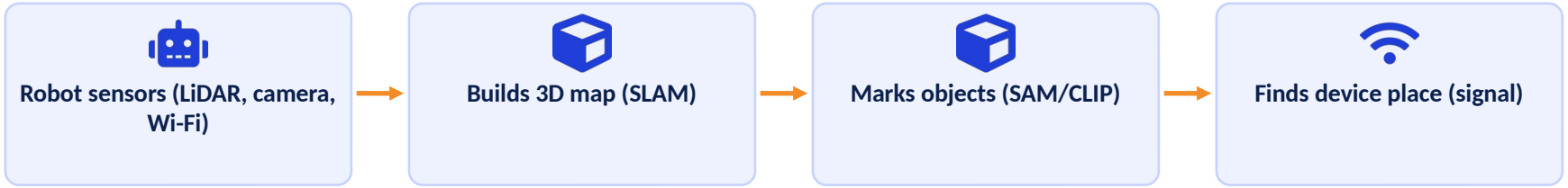
Point	Before	SpectraTwin
How we know a device	IP or MAC address	Habits + place on 3D map
Address changes	We lose the device	We still match it
Fake device	Hard to catch	We check if looks and acts match
Privacy	Fixed IDs, or nothing	Hidden IDs, rough place, no people
Labels	Need hand labels	No labels needed

We will measure: match accuracy (with and without place), wrong matches, fake device catch rate, privacy risk.

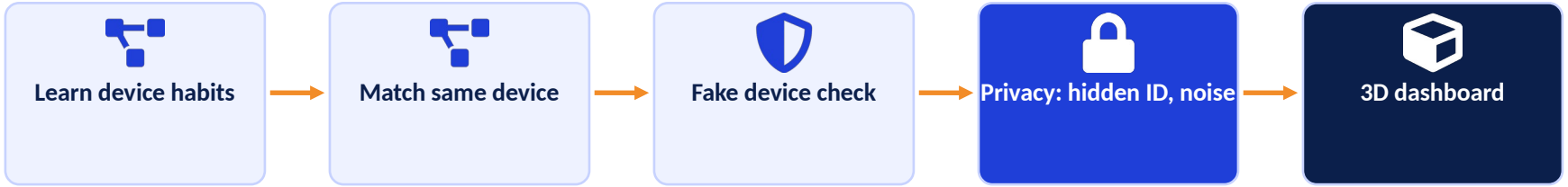
Architecture

Part A: robot sees. Part B: we match, protect and show.

Part A: Robot sees



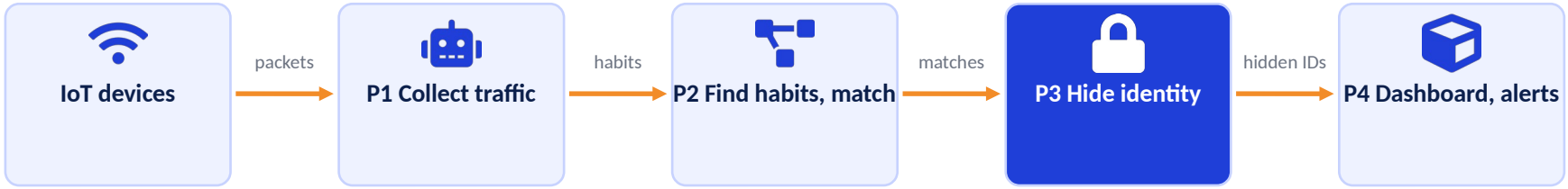
Part B: Match, protect, show



We test in a simulator (Gazebo) with public IoT data. We also plan a small real test.

Data Flow

How data moves, and where we hide identity



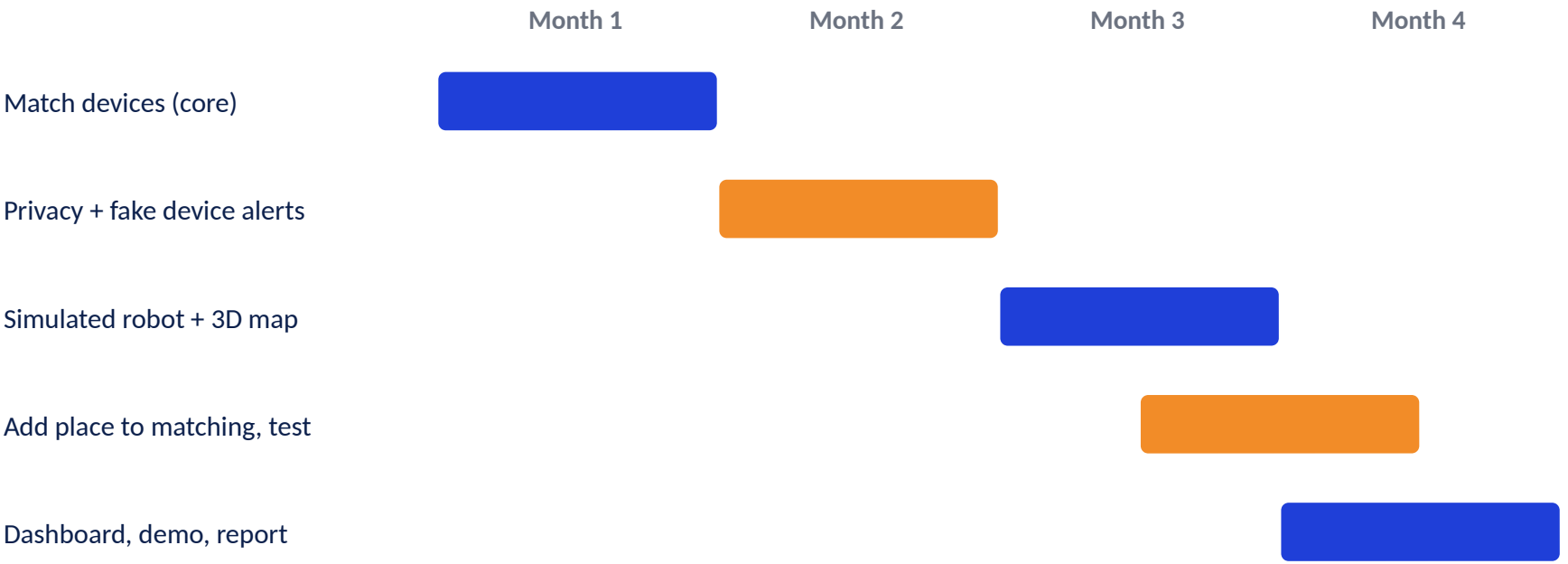
D1 Hidden ID store

D2 3D map (objects only, no people)

Real IP addresses stay inside P2. Only hidden IDs and rough places are saved.

Timeline

4 months. We test each step before the next one.



We use public data (UNSW IoT, CICIoT2023) and fake address changes. We test with and without place.

Bibliography

- [1] T. Narten, R. Draves, S. Krishnan, "Privacy Extensions for SLAAC in IPv6," RFC 4941, 2007; replaced by F. Gont et al., RFC 8981, 2021.
- [2] F. Gont, "Security and Privacy Considerations for IPv6 Address Generation Mechanisms," RFC 7721, 2016.
- [3] A. Sivanathan et al., "Classifying IoT Devices in Smart Environments Using Network Traffic Characteristics," IEEE TMC, 2019.
- [4] T. Chen et al., "A Simple Framework for Contrastive Learning of Visual Representations," ICML, 2020.
- [5] W. Xu et al., "FAST-LIO2: Fast Direct LiDAR-Inertial Odometry," IEEE T-RO, 2022.
- [6] A. Kirillov et al., "Segment Anything," ICCV, 2023.
- [7] E. Neto et al., "CICIoT2023: A Real-Time Dataset and Benchmark for Large-Scale Attacks in IoT Environment," Sensors, 2023.

Thank You!

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